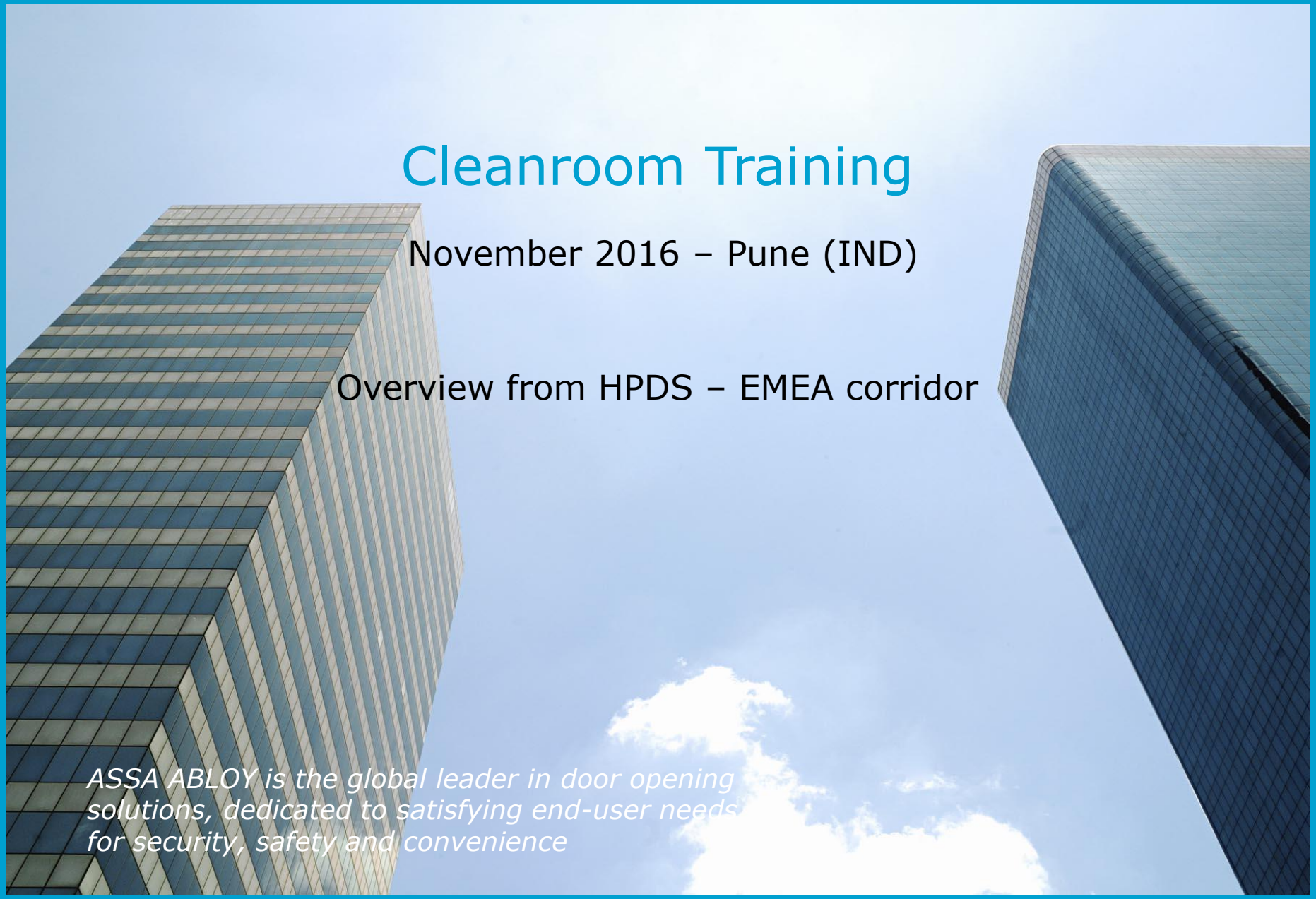


Cleanroom Training

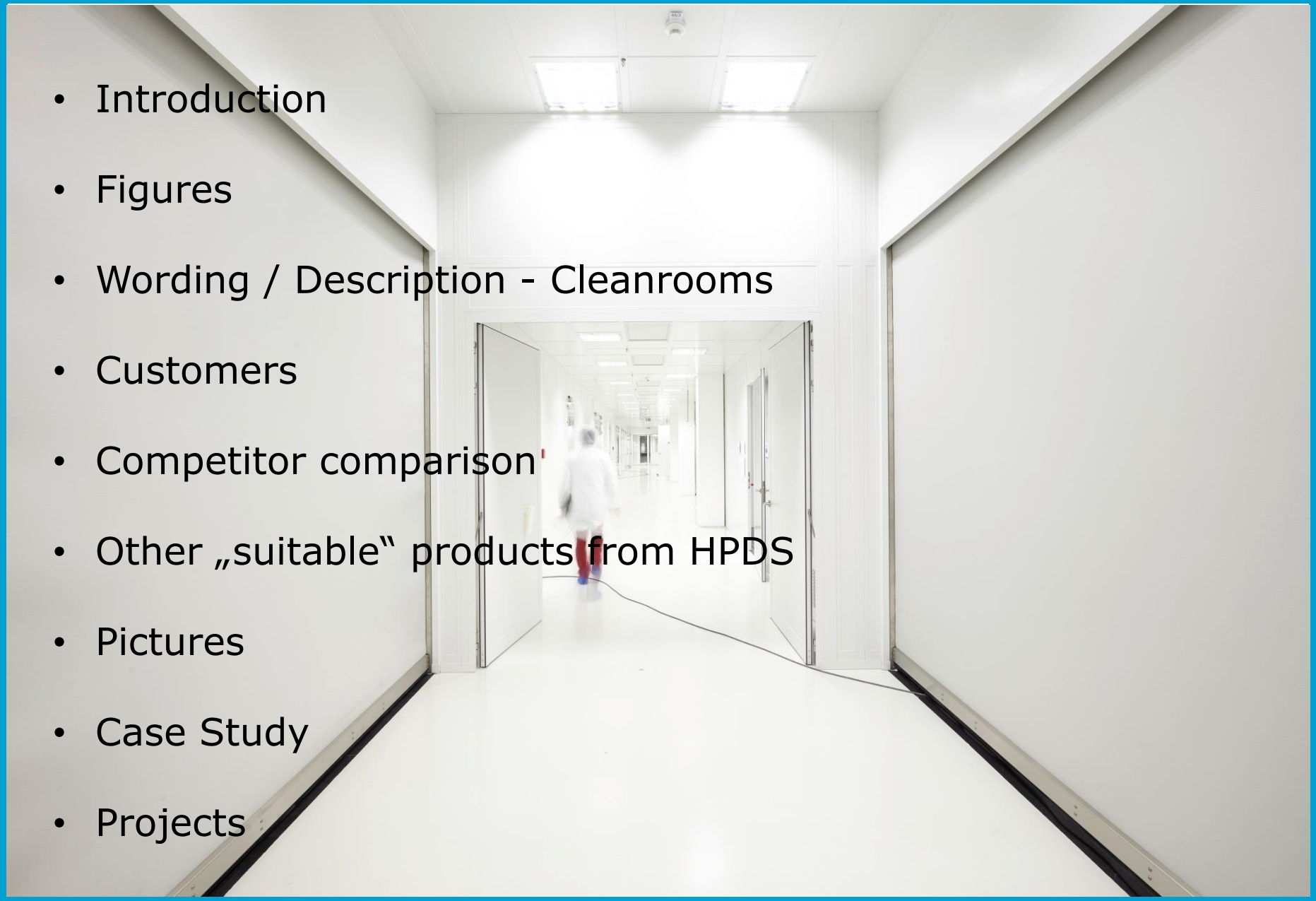
November 2016 – Pune (IND)

Overview from HPDS – EMEA corridor

ASSA ABLOY is the global leader in door opening solutions, dedicated to satisfying end-user needs for security, safety and convenience

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- Introduction
- Figures
- Wording / Description - Cleanrooms
- Customers
- Competitor comparison
- Other „suitable“ products from HPDS
- Pictures
- Case Study
- Projects



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ASSA ABLOY

Daniel Grüttner-Mierswa
Key Account Manager EMEA - Clean Room Doors
High Performance Door Solutions

ASSA ABLOY Entrance Systems
Albany Door Systems GmbH, Am Mondschein 25, DE - 59557 Lippstadt
Direct: +49 (0) 29 41-766-644 Mobile: +49 (0) 160 - 94 68 71 55
Switchboard: +49 (0) 29 41-766-0 Fax: +49 (0) 2 31 - 9 34 90 69
daniel.gruettner-mierswa@assaabloy.com
www.assaabloyentrance.com

Figures (HPDS EMEA – RR300 Clean)

Turnover 2015

- 2.18 Mio € (320pcs)
- +17 to Budget (1.8 Mio €)
- +24% to 2014 (turnover)

Quotes 2015

- Doors 837pcs
- Volume 6.3 Mio €
- Hitrate 33% (must be at ~40%)

Turnover 2016 (Jan-Oct)

- 1.55 Mio € (215pcs)
- Estimated ~ 1.95 Mio € (270pcs)

Quotes 2016 (Jan-Oct)

- Doors 925pcs
- Volume 7.8 Mio €

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What is a cleanroom ?

A Cleanroom is a room where the concentration of airborne particles is being kept as less as needed.

(Quelle: Wikipedia)

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How to do this ?

By establish a controlled environment with a kind of air condition using special filters and over- or underpressure.

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Why do cleanrooms - controlled environment exist ?

To protect sensitive products/processes against contamination and partially the environment against the product.

AND

due to customer requests !

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Classification of Cleanrooms

Pharmaceutical Cleanrooms

- EU-GMP Guide
- focus on particle concentration AND design (easy to clean / germ forming units)
- GMP-A highest
  GMP-D lowest

Technical Cleanrooms

- DIN EN ISO 14644-1
- Focus „only" on particle concentration
- ISO 1 highest
  ISO 9 lowest

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Classification of Cleanrooms

Pharmaceutical Cleanrooms

Reinraumklassen nach GMP-Leitfaden Annex 1[2]

	Maximal erlaubte Partikelzahl pro m^3			
Klasse	Ruhezustand ≥ 0,5 µm	Ruhezustand ≥ 5 µm	Betriebszustand ≥ 0,5 µm	Betriebszustand ≥ 5 µm
A	3.520	20	3.250	20
B	3.520	29	352.000	2.900
C	352.000	2.900	3.520.000	29.000
D	3.520.000	29.000	nicht festgelegt	nicht festgelegt

Klasse / Grade	Empfohlene Genzwerte für die mikrobiologische Kontamination (a) / Recommended limits for microbial contamination (a)			
	Luftprobe KBE/m^3	Sedimentations-platten (Durchm. 90 mm) KBE/4 Stunden (b)	Kontaktplatten (Durchm. 55 mm) KBE/Platte	Handschuh-abdruck 5 Finger KBE/Handschuh
	air sample cfu/m^3	settle plates (diam. 90 mm) cfu/4 hours (b)	contact plates (diam. 55 mm) cfu/plate	glove print 5 fingers cfu/glove
A	<1	<1	<1	<1
B	10	5	5	5
C	100	50	25	–
D	200	100	50	–

Technical Cleanrooms

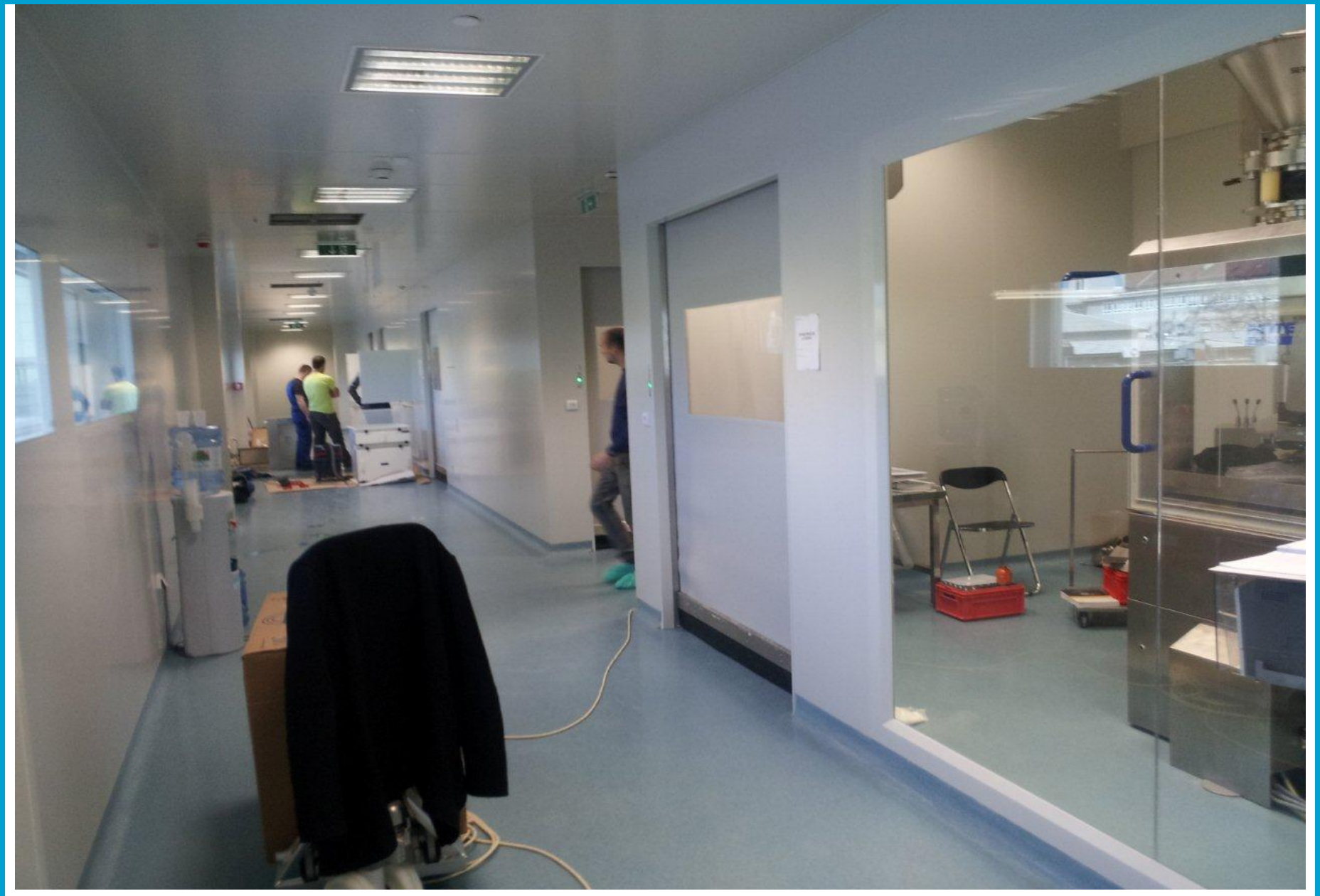
Reinraumklassen nach ISO 14644-1[2]

	Partikel je m^3					
Klasse	≥ 0,1 µm	≥ 0,2 µm	≥ 0,3 µm	≥ 0,5 µm	≥ 1,0 µm	≥ 5,0 µm
ISO 1	10	2				
ISO 2	100	24	10	4		
ISO 3	1.000	237	102	35	8	
ISO 4	10.000	2.370	1.020	352	83	
ISO 5	100.000	23.700	10.200	3.520	832	29
ISO 6	1.000.000	237.000	102.000	35.200	8.320	293
ISO 7				352.000	83.200	2.930
ISO 8				3.520.000	832.000	29.300
ISO 9				35.200.000	8.320.000	293.000

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Who are our customers !?

- **End User** (Pharmeutical, Semiconductor, Bio-Technic, Life-Science, Optics, Mikroelectronics, Aerospace, Cosmetics, Medicine-Technic…)

- **Architects- and Engineering offices** (Carpus+Partner, M+W, Chemgineering, NNE-Pharmaplan, Jacobs, K2, io consultants, HWP,…)

- **Cleanroom construction companies** (Ritterwand, Lindner, Daldrop, Ardmac, Asgard, Telstar, Renex, Citea, Parteco, Cleangrad, …)

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- **End User** (Pharmeutical, Semiconductor, Bio-Technic, Life-Science, Optics, Mikroelectronics, Aerospace, Cosmetics, Medicine-Technic…)

  - Small projects (~ 5 Mio. € ?!?)
  - Engineering department
  - Replacements
  - Facility Management
  - Choose existing suppliers
  - Sometimes hard to convince

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- **Architects- and Engineering offices** (Carpus+Partner, M+W, Chemgineering, NNE-Pharmaplan, Jacobs, K2, io consultants, HWP,…)

  - Big Projects – Green Field
  - Complete rebuilding
  - Sometimes „Turnkey" projects
  - Strong view on guidelines
  - Planning reliability
  - Relationship
  - Support

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- **Cleanroom construction companies** (Ritterwand, Lindner, Daldrop, Ardmac, Asgard, Telstar, Renex, Citea, Parteco, Cleangrad, …)

  - Price
  - Integration in their system
  - Support
  - Flexible and on schedule
  - Partnership

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EFAFLEX
schnelle und sichere Tore



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schnelle und sichere Tore

Fraunhofer
IPA
TESTED® DEVICE
Albany Door Systems
RapidRoll Cleanroom
Report No. AL 1003-514

Certified through the Fraunhofer Institute IPA in accordance to DIN EN ISO 14644-1 **AND** GMP.

Baumuster geprüft
Produktion überwacht
TÜV SÜD
DIN EN ISO 14644
VDI 2083

Certified through the TÜV South Cleancert **ONLY** in accordance to DIN EN ISO 14644-1

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EFAFLEX
schnelle und sichere Tore

Opening up to 2,0 m/s
and
Closing up to 1,0 m/s

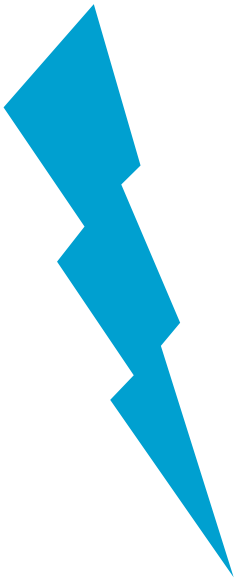
Opening up to 1,0 m/s
and
Closing up to 0,5 m/s

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schnelle und sichere Tore

Choice between contact edge (standard) or pre-running photocell (option - patent)



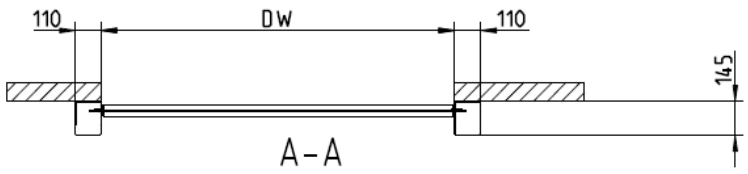
Only radio transmitted contact edge
(life cycle of batteries; interruption of transmission)

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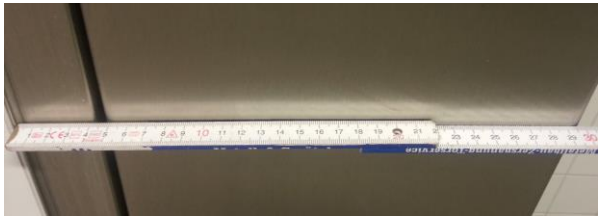
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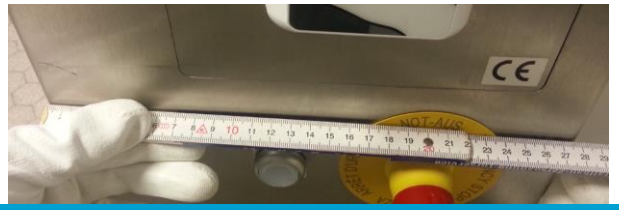
EFAFLEX
schnelle und sichere Tore

Side columns only 110mm width, 145mm deep.
(overall less space needed)



Side columns up to 2,5 times deeper.
(overall big space needed)



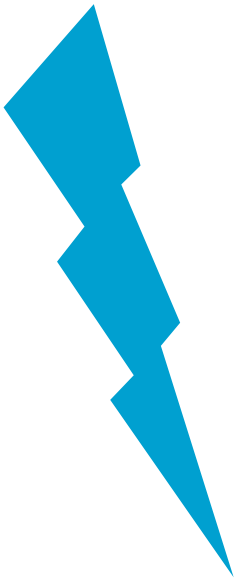


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EFAFLEX
schnelle und sichere Tore

Emergency egress with hand crank or very comfortable UPS
(Status via LED, maintenance free, Life-cycle 10 years, opening and closing up to 50 times, environment stay controlled)

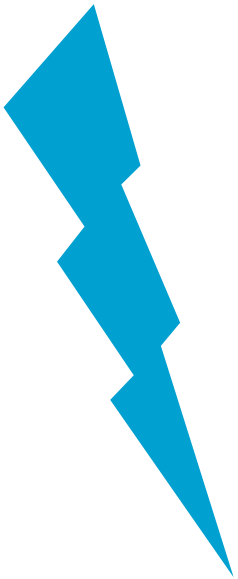


Emergency egress with tensions via pushbutton
(opening only up to the half height, maintenance – wear parts, particles)

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EFAFLEX
schnelle und sichere Tore



Max. dimensions
3500 x 3500 mm
(min. 1000x1000mm)

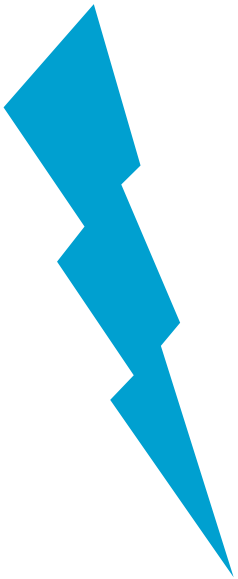
Max. dimensions
2500 x 3000 mm
(min. ???)

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EFAFLEX
schnelle und sichere Tore

Entire door frame and covers in SS (304)or galvanized steel
(powder coated / 316 possible with Ref.)



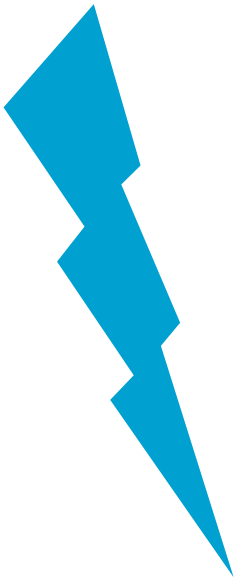
Entire door frame and covers in SS (304)

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EFAFLEX
schnelle und sichere Tore

PVC-curtain with vertically reinforcement stripes
(biggest transparency, various colours, antistatic/FDA normally not required)



Complete closed Curtain (PE – antistatic, FDA compliant), horizontal window available.
(window is PVC – not antistatic anymore.)

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EFAFLEX
schnelle und sichere Tore

Integrated traffic light:
red, green, yellow
flashing and MagicSwitch
(flush)





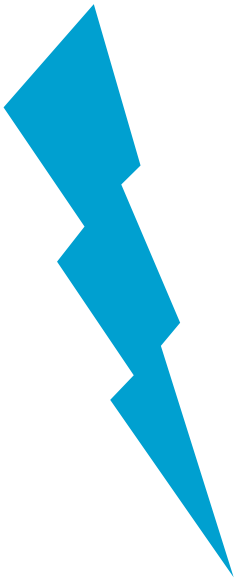
Integrated traffic light:
red, green and MagicSwitch
(circumferential edge)





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MCC – control system
(developed by ourselves, different BUS-Modules available)



EFA-CON – control system
(supplier: FEIG; bulk product - same like Hörmann)

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And the winner is…

ASSA ABLOY Entrance Systems

besam ASSA ABLOY | Crawford ASSA ABLOY | ALBANY ASSA ABLOY | MEGADOOR ASSA ABLOY

with the High Performance Door Albany RR300 Clean

(fast, safe, compact, variable, almost no wear parts)

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RR300 Plus Alu - Sealed:

- suitable up to ISO class 7
- Leckage at dimension 1500x2000mm and 25Pa – 65m³/h
- available up to 3000x3000mm (W/H)
- door construction made out of anodized Alu (motor cover plastic)
- manual self opening with hand lever via counter balancing system
- PVC door curtain with reinforcement stripes
- stationary photocell and safety contact edge
- MCC under drive unit
- Display on side frame or wall



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SP Swedish National Testing and Research Institute

Test results

Air permeability

Leakage at 50 Pa positive pressure:	7.6 $m^3/h,m^2$
Leakage at 50 Pa negative pressure:	5.5 $m^3/h,m^2$

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https://www.youtube.com/watch?v=PVKaPUD8JWM

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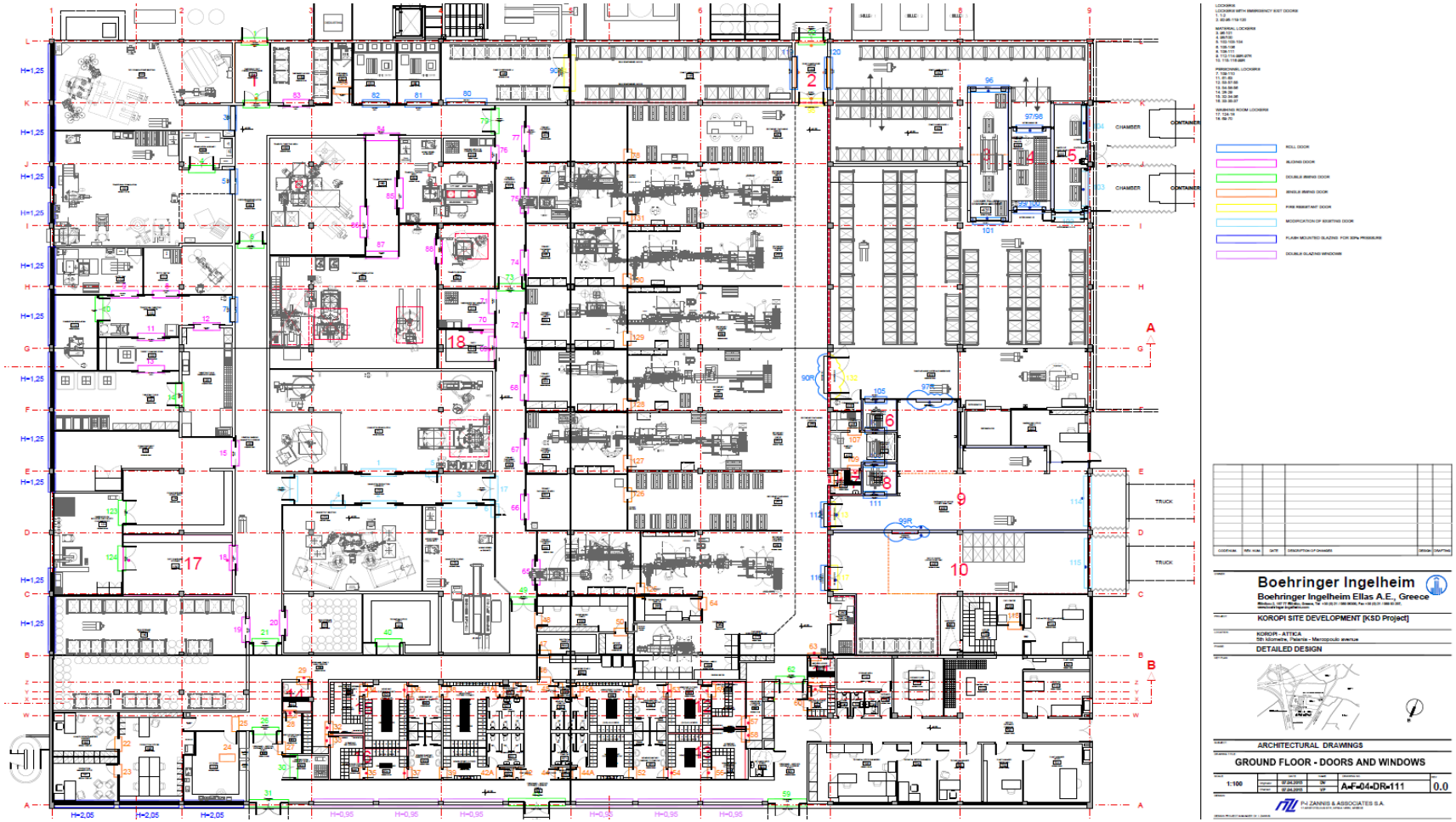
How things are going
Case Study – Böhringer Ingelheim Greece

The Boehringer Ingelheim group is one of the world's 20 leading pharmaceutical companies. Headquartered in Ingelheim, Germany, Boehringer Ingelheim operates globally with 146 affiliates and a total of more than 47,700 employees. The focus of the family-owned company, founded in 1885, is researching, developing, manufacturing and marketing new medications of high therapeutic value for human and veterinary medicine. In 2014, Boehringer Ingelheim achieved net sales of about 13.3 billion euros. R&D expenditure corresponds to 19.9% of its net sales. Boehringer Ingelheim's successes in research & development continuously strengthen its portfolio of medications and offer patients true therapeutic benefit. The company's research focuses on four areas of strength: Immunology and Respiratory, Cardiometabolic diseases, Central Nervous System diseases, and Oncology. With its research functions "Scientific Platforms" and "Research Beyond Borders" Boehringer Ingelheim is prepared to leverage synergies by investigating scientific concepts with potential in more than one disease area and by exploring emerging science and technology both within and outside the current research therapeutic focus of Boehringer Ingelheim.

How things are going
Case Study – Böhringer Ingelheim Greece

- First knowledge of project in November 2014
  - Inquiry from a german cleanroom builder
  - Formless, 10 to 30 doors – 4 different dimension
- First spec´s – tender, drawings, layouts we´ve got in April 2015
  - Clarification; Discussions about required and useful solutions
- Common visit at site together with Lindner, PDS + HPDS in End of May 2015

How things are going
Case Study – Böhringer Ingelheim Greece



How things are going
Case Study – Böhringer Ingelheim Greece

- Conclusion of visit:
  - Complete refurbishement
  - Realiziation in different steps (until end of 2015)
  - 18 HPD; modification of 3 existing doors from IDDS; 27 sliding doors; couple of door openers for „normal“ swing doors
  - Competitors on site: Efaflex and Novoferm
  - To do: spread a lot of information; revision of our offer.

How things are going
Case Study – Böhringer Ingelheim Greece

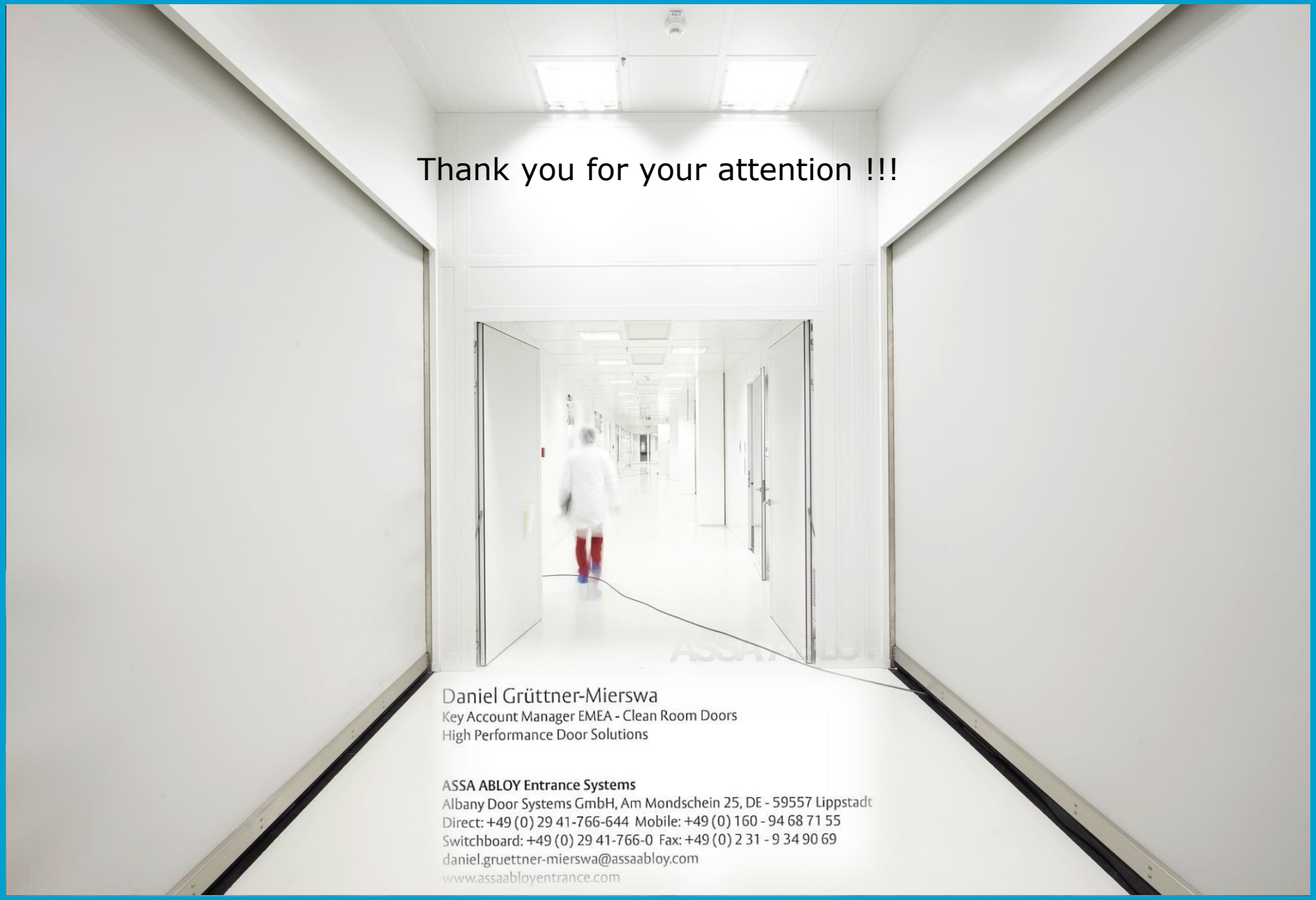
- Headquarter of BI in Germany made the decision to give the design contract to an external architect (August 2015)
- September 2016 we´ve got next inquiry from a second cleanroom builder (Daldrop also GER).
  - Fighting on 4 fronts (Lindner, Daldrop, BI, architect)
  - All parties involved favorite HPDS but for sure we were in competition to Efaflex

How things are going
Case Study – Böhringer Ingelheim Greece

- End of November 2016 we signed a contract with Daldrop about 20pcs of RR300 Clean (~150.000 €).
  - 10pcs should be delivered and installed until mid of January
    - These 10pcs were delivered in the beginning of January but only 3pcs were installed in March (with supervisor from HPDS)!
    - Other 7pcs are still waiting on site of the installation
  - When we can deliver the remaining 10 doors is not in sight
  - In the Meantime 1 door was hit by a forklift. Together with some additional work it was ordered directly at our partner in Greece – standoor.

Latest Projects

- Roche in Mannheim – 27 RR300Clean / 180.000 €
- Bayer in Wuppertal – 11 RR300Clean / 110.000 €
- Bayer in Leverkusen – 38 (Clean – 300Plus F+R)250.000 €
- Pfizer in Freiburg – 8 RR300Clean / 80.000 €
- Novartis (CH) – 36 RR300 Clean / 210.000 €
- Böhringer (GR) – 20 RR300Clean / 150.000 €
- Novo Nordisk (DK) – 15 RR300Cleaan / 105.000 €
- Grotex (RUS) – 20 RR300Clean / 130.000 €
- …



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